

Research Question

How does ambient lighting influence heart rate variability (RMSSD), heart rate (HR), and cognitive task performance during a mental arithmetic task?

Introduction

Ambient lighting is an important environmental factor that can influence human comfort, attention, and cognitive performance. Previous research suggests that lighting conditions may affect alertness and mental workload, but their effects on physiological responses during cognitive tasks remain unclear. Heart rate (HR) and heart rate variability (HRV), particularly the Root Mean Square of Successive Differences (RMSSD), are widely used indicators of autonomic nervous system activity and mental workload. This study investigated whether bright and dim lighting influence heart rate, heart rate variability (RMSSD), and arithmetic task performance during a mental arithmetic task using a Polar H10 ECG sensor in a within-subject experimental design.

Method

- Participants:** 20 healthy university students.
- Design:** Within-subject experiment (each participant completed both lighting conditions).
- Conditions:** Bright lighting and Dim lighting (counterbalanced order).
- Task:** 5-minute mental arithmetic task under each lighting condition.
- Sensor:** Polar H10 ECG chest strap.
- Measurements:** Heart Rate (HR), Heart Rate Variability (RMSSD), and arithmetic task accuracy.
- Data Processing:** ECG signals processed in Python using NeuroKit2 to extract HR and RMSSD.
- Statistical Analysis:** Shapiro–Wilk normality test, paired t-test (HR & RMSSD), Wilcoxon signed-rank test (accuracy), significance level $\alpha = 0.05$.

Result

Heart Rate Comparison
Figure 1. Heart Rate (HR) under Bright and Dim Lighting

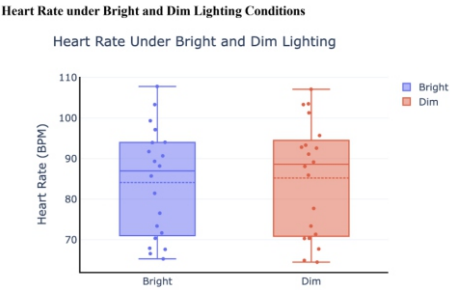
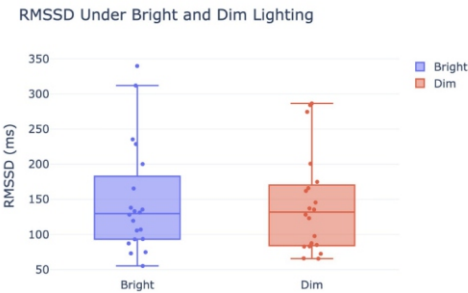


Figure 1. Boxplots showing the distribution of heart rate (HR) measured under bright and dim lighting conditions. Each point represents an individual participant.

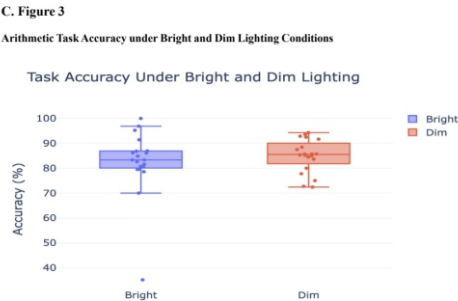
Finding: Mean HR was slightly different between conditions, but the difference was not statistically significant ($p = 0.130$).

Heart Rate Variability (RMSSD)
Figure 2. RMSSD under Bright and Dim Lighting



Finding: RMSSD showed no significant difference between lighting conditions ($p = 0.560$).

Cognitive Performance
Figure 3. Arithmetic Task Accuracy under Bright and Dim Lighting



Finding: Arithmetic task accuracy was similar in both lighting conditions ($p = 0.388$).

Statistical Summary

Variable	Bright Lighting Mean $\pm$ SD	Dim Lighting Mean $\pm$ SD	Statistical Test	p-value
Heart Rate (HR)	84.10 $\pm$ 13.24 bpm	85.20 $\pm$ 13.61 bpm	Paired t-test	0.130
RMSSD	147.85 $\pm$ 74.38 ms	142.94 $\pm$ 72.27 ms	Paired t-test	0.560
Task Accuracy	82.47 $\pm$ 12.50 %	84.98 $\pm$ 13.47 %	Wilcoxon Signed-Rank Test	0.388

Statistical comparison of heart rate (HR), heart rate variability (RMSSD), and arithmetic task accuracy under bright and dim lighting conditions. The table presents the mean $\pm$ standard deviation for each variable together with the corresponding statistical test and p-value. These results summarize the physiological and cognitive responses observed across all participants under the two experimental lighting conditions.

Conclusion

This study investigated the effect of ambient lighting on physiological responses and cognitive performance during a mental arithmetic task. The results showed no statistically significant differences in heart rate (HR), heart rate variability (RMSSD), or arithmetic task accuracy between the bright and dim lighting conditions. These findings suggest that short-term changes in ambient lighting had minimal influence on autonomic activity and cognitive performance under the experimental conditions. Future research with larger sample sizes, longer exposure durations, and additional physiological measures may provide a more comprehensive understanding.